1. Low Spacial Resolution - Sampling

The problem is insufficient sampling, where the camera captures too few pixels to represent small objects clearly. According to the sampling concept, higher spacial sampling gives more pixels and better image detail.

2. Dark and Noisy Night image - Enhancement

The image can be improved using histogram equalization and noise filtering. Histogram equalization increases the contrast of dark regions and make objects more visible. A suitable noise filter, such as a median filter, can reduce unwanted noise

3. Continuous Intensity values - Quantization

Quantization converts continuous intensity values into a finite number of digital intensity levels. For ex, an image can be represented using 256 intensity levels. Each continuous intensity value is mapped to the nearest available digital level.

4. Uneven lighting - Image Enhancement

Contrast enhancement can be applied to improve the visibility of product defects. Techniques such as adaptive histogram equalization (CLAHE) enhance local contrast

It makes dark and bright regions more balanced despite uneven illumination.

Therefore, small defects become easier for the inspection system to identify

5. Roads, Vechicles and Pedestrians – Image Segmentation.

Image segmentation divides the image into meaningful regions such as roads, Vechiles and pedestrians. Semantic or instance segmentation can assign different labels to these objects. The self-driving car can thus understand the position and boundaries of each object.

6. Crack Detection – Edge detection:

Edge detection such as the canny edge detector, can be applied to the captured image. It identifies sudden changes in intensity corresponding to boundaries and cracks.

The detected edges can highlight the shape and detection / location of cracks on components. This helps automate defect detection and improve

7. Noisy Medical Image – filtering

A median filter can be used to remove unwanted noise while preserving important edges

It is particularly suitable for removing salt and pepper noise in medical images. After filtering the image becomes smoother and the tumor region can be more clearly observed. This improves the visibility of important structures for further analysis

8. Vechile Detection & Tracking:

Object detection can identify vechiles by drawing bounding boxes around them.

A tracking algorithm can then follow each detected vechile across consecutive video frames. This provides information about vechile count, position, movement and traffic flow. It helps in traffic monitoring congestion analysis and intelligent traffic management.

9. Motion Blur:

The blur is mainly caused by object or camera movement during image acquisition, especially with a slow shutter speed. A suitable technique is image deblurring, such as inverse filtering or Wiener filtering It attempts to restore sharp details that were lost because of motion blur.

Using a faster shutter speed during capture can also reduce the position.

(b). Defective vs Good products — Image classification.

Image classification assigns captured product images to classes such as "defective" or "good".

A trained machine-learning or deep-learning model can learn visual features of defects.

Each new product image is classified automatically based on its appearance. This supports fast, consistent, and automated quality inspection